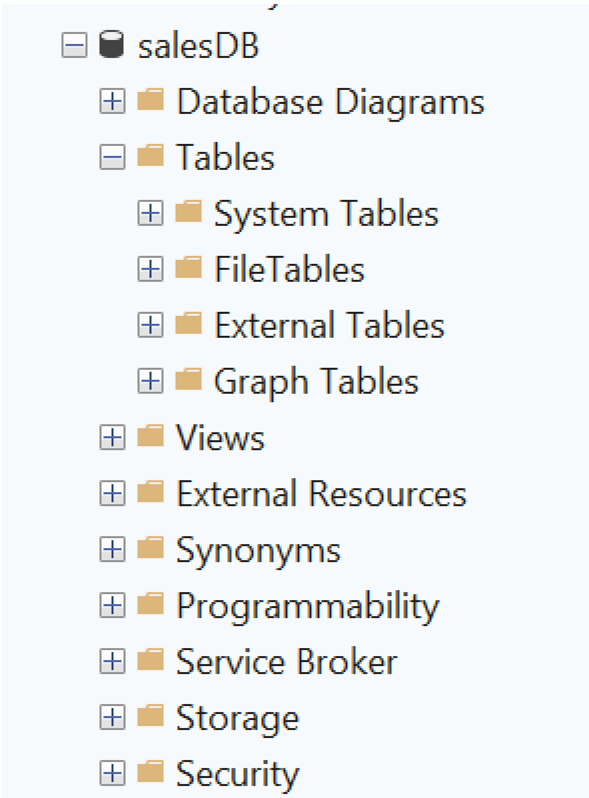
Weekly Assignment-1

2/1/26

create database salesDB;



use salesDB;

Create client_master table

```
create table client_master (
 clientno varchar(6) primary key
  constraint ck_clientno check (clientno like 'C%'),
 name varchar(20) not null,
 address1 varchar(30),
 address2 varchar(30),
 city varchar(15),
 pincode int,
```

```
  state varchar(15),
  baldue decimal(10,2)
);
```

Create product_master table

```
create table product_master (
  productno varchar(6) primary key
    constraint ck_productno check (productno like 'P%'),
  description varchar(15) not null,
  profitperc decimal(4,2) not null,
  unitmeasure varchar(10) not null,
  qtyonhand int not null,
  reorderlvl int not null,
  sellprice decimal(8,2)
    constraint ck_sellprice check (sellprice <> 0),
  costprice decimal(8,2)
    constraint ck_costprice check (costprice <> 0)
);
```

Create salesman_master table

```
create table salesman_master (
  salesmanno varchar(6) primary key
    constraint ck_salesmanno check (salesmanno like 'S%'),
  salesmanname varchar(20) not null,
  address1 varchar(30) not null,
  address2 varchar(30),
```

```
  city varchar(20),
  pincode int,
  state varchar(20),
  salamt decimal(8,2)
    constraint ck_salamt check (salamt <> 0),
  tgttoget decimal(6,2) not null,
  ytdsales decimal(6,2) not null,
  remarks varchar(60)
);
```

Create sales_order table

```
create table sales_order (
  orderno varchar(6) primary key
    constraint ck_orderno check (orderno like 'O%'),
  clientno varchar(6) references client_master(clientno),
  orderdate date,
  delyaddr varchar(25),
  salesmanno varchar(6) references salesman_master(salesmanno),
  delytype char(1)
    constraint ck_delytype check (delytype in ('P','F')),
  billedyn char(1)
    constraint ck_billedyn check (billedyn in ('Y','N')),
  delydate date,
  orderstatus varchar(15)
    constraint ck_orderstatus check
    (orderstatus in ('In Process','Fulfilled','Backorder','Cancelled'))
```

```
);
```

Create sales_order_details table

```
create table sales_order_details (
  orderno varchar(6) references sales_order(orderno),
  productno varchar(6) references product_master(productno),
  qtyordered int,
  qtydisp int,
  productrate decimal(10,2),
  constraint pk_order_details primary key (orderno, productno)
);
```

Insert values in client_master table

```
insert into client_master values
('C0001','Ivan Bayross','A-12 MG Road',null,'Mumbai',400001,'Maharashtra',1200.00),
('C0002','Suresh','B-15 FC Road',null,'Pune',411001,'Maharashtra',800.00),
('C0003','Anita','C-22 Park St',null,'Mumbai',400002,'Maharashtra',1500.00),
('C0004','Kiran','D-10 Ring Rd',null,'Delhi',110001,'Delhi',500.00),
('C0005','Neha','E-18 Lake Rd',null,'Chennai',600001,'Tamil Nadu',2000.00);
```

Insert values in product_master table

```
insert into product_master values
('P0001','Trousers',20.00,'Nos',50,10,2500.00,2000.00),
('P0002','Pull Overs',15.00,'Nos',5,5,1500.00,1200.00),
('P0003','Shirts',10.00,'Nos',100,20,1800.00,1400.00),
('P0004','Jackets',25.00,'Nos',2,5,4500.00,3500.00),
```

('P0005','Socks',5.00,'Pairs',500,50,300.00,200.00);

Insert values in salesman_master table

insert into salesman_master values

('S0001','Aman','Hill Rd',null,'Mumbai',400003,'Maharashtra',5000.00,100.00,50.00,'Good'),

('S0002','Ravi','Link Rd',null,'Pune',411002,'Maharashtra',6000.00,150.00,120.00,'Excellent'),

('S0003','John','MG Rd',null,'Delhi',110002,'Delhi',4500.00,80.00,30.00,'Average'),

('S0004','Sara','Park St',null,'Mumbai',400004,'Maharashtra',7000.00,200.00,180.00,'Top'),

('S0005','Kiran','Beach Rd',null,'Chennai',600002,'Tamil Nadu',4000.00,70.00,40.00,'New');

Insert values in sales_order table

insert into sales_order values

('O19001','C0001','2022-04-10','Mumbai','S0001','P','Y','2022-04-15','Fulfilled'),

('O19002','C0002','2022-05-05','Pune','S0002','F','N','2022-05-10','In Process'),

('O19003','C0003','2022-06-01','Mumbai','S0004','P','Y','2022-06-06','Fulfilled'),

('O19004','C0001','2022-03-15','Mumbai','S0001','F','Y','2022-03-20','Cancelled'),

('O19005','C0004','2022-07-12','Delhi','S0003','P','N','2022-07-18','Backorder');

Insert values in sales_order_details table

insert into sales_order_details values

('O19001','P0001',3,3,2500.00),

('O19001','P0002',2,2,1500.00),

('O19002','P0003',10,8,1800.00),

('O19003','P0005',100,100,300.00),

('O19004','P0002',1,1,1500.00);

Answer following queries with the help of above schema

1. Display the names of all the clients.

select name ClientName from client_master;

	ClientName
1	Ivan Bayross
2	Suresh
3	Anita
4	Kiran
5	Arjun

2. Display all the clients who are located in Mumbai.

select * from client_master where city='Mumbai';

	clientno	name	address1	address2	city	state	baldue	phone_no
1	C0001	Ivan Bayross	A-12 MG Road	NULL	Mumbai	Maharashtra	1200.00	NULL
2	C0003	Anita	C-22 Park St	NULL	Mumbai	Maharashtra	1500.00	NULL

3. Display all the products whose selling price is > 2000 and < 5000.

select * from product_master where sellprice>2000 and sellprice<5000;

	productno	description	profitperc	unitmeasure	qtyonhand	reorderlvl	sellprice	costprice
1	P0001	Trousers	20.00	Nos	50	10	2500.00	2000.00
2	P0004	Jackets	25.00	Nos	2	5	4500.00	3500.00

4. Display Name, City and State of Clients not in the state of Maharashtra.

select name,city,state from client_master where state not in ('Maharashtra');

	name	city	state
1	Kiran	Delhi	Delhi
2	Neha	Chennai	Tamil Nadu

5. Display Name, City and State of Clients not in the state of Maharashtra.

select * from client_master where clientno in ('C0001','C0002');

	clientno	name	address1	address2	city	state	baldue	phone_no
1	C0001	Ivan Bayross	A-12 MG Road	NULL	Mumbai	Maharashtra	1200.00	NULL
2	C0002	Suresh	B-15 FC Road	NULL	Pune	Maharashtra	800.00	NULL

6. Change the selling price of '1.44 drive' to Rs. 1150.50.

update product_master set sellprice = 1150.50 where description = '1.44 drive';

select * from product_master where description = '1.44 drive';

	productno	description	profitperc	unitmeasure	qtyonhand	reorderlvl	sellprice	costprice
1	P0010	1.44 drive	10.00	Nos	20	5	1150.50	800.00

7. Delete the record of client_no C0005.

delete from client_master where clientno='C0005';

select clientno from client_master;

	clientno
1	C0001
2	C0002
3	C0003
4	C0004

8. Display the clients who stay in a city whose second letter is 'a'.

select * from client_master where city like '_a%';

	clientno	name	address1	address2	city	pincode	state	baldue
1	C0006	Arjun	Main Road	NULL	Nagpur	440001	Maharashtra	900.00

9. Count the number of products having price greater than or equal to 1500.

select * from product_master where sellprice>=1500;

	productno	description	profitperc	unitmeasure	qtyonhand	reorderlvl	sellprice	costprice
1	P0001	Trousers	20.00	Nos	50	10	2500.00	2000.00
2	P0002	Pull Overs	15.00	Nos	5	5	1500.00	1200.00
3	P0003	Shirts	10.00	Nos	100	20	1800.00	1400.00
4	P0004	Jackets	25.00	Nos	2	5	4500.00	3500.00

10. Display qtyordered, qtydisp and balancedqty (not in table).

select qtyordered, qtydisp, qtyordered - qtydisp as balancedqty from sales_order_details;

	qtyordered	qtydisp	balancedqty
1	3	3	0
2	2	2	0
3	10	8	2
4	100	100	0
5	1	1	0

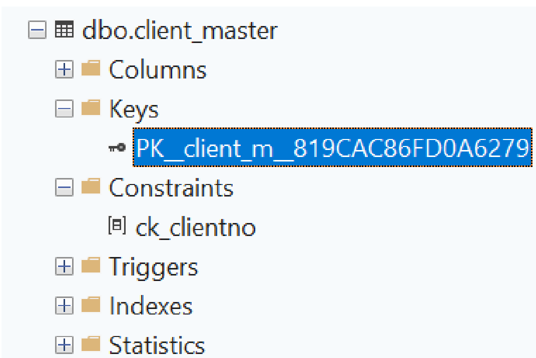
Write Commands to do following

1. Make Client_no as primary key in client_master.

alter table client_master

add constraint ck_clientno

primary key(clientno);



2. Add a new column phone_no in the client_master table.

alter table client_master add phone_no varchar(15);

select * from client_master;

	clientno	name	address1	address2	city	pincode	state	baldue	phone_no
1	C0001	Ramesh	A-12 MG Road	NULL	Mumbai	400001	Maharashtra	1200.00	NULL
2	C0002	Suresh	B-15 FC Road	NULL	Pune	411001	Maharashtra	800.00	NULL
3	C0003	Anita	C-22 Park St	NULL	Mumbai	400002	Maharashtra	1500.00	NULL
4	C0004	Kiran	D-10 Ring Rd	NULL	Delhi	110001	Delhi	500.00	NULL
5	C0006	Arjun	Main Road	NULL	Nagpur	440001	Maharashtra	900.00	NULL

3. Add the not null constraint in the product_master table with the column description, profit percent, sell price and cost price.

--description column

alter table product_master

alter column description varchar(15) not null;

--profitperc column

alter table product_master

alter column profitperc decimal(4,2) not null;

--sellprice column

alter table product_master

alter column sellprice decimal(8,2) not null;

--costprice column

alter table product_master

alter column costprice decimal(8,2) not null;

```
Commands completed successfully.

Completion time: 2026-01-04T12:33:14.9739624+05:30
```

4. Change size of name column to 60 in client_master table.

alter table client_master

alter column name varchar(60);

Column Name	Data Type
clientno	varchar(6)
name	varchar(60)
address1	varchar(30)
address2	varchar(30)
city	varchar(15)
pincode	int
state	varchar(15)
baldue	decimal(10, 2)
phone_no	varchar(15)

5. Remove pincode column from table.

alter table client_master

drop column pincode;

select * from client_master;

	clientno	name	address1	address2	city	state	baldue	phone_no
1	C0001	Ivan Bayross	A-12 MG Road	NULL	Mumbai	Maharashtra	1200.00	NULL
2	C0002	Suresh	B-15 FC Road	NULL	Pune	Maharashtra	800.00	NULL
3	C0003	Anita	C-22 Park St	NULL	Mumbai	Maharashtra	1500.00	NULL
4	C0004	Kiran	D-10 Ring Rd	NULL	Delhi	Delhi	500.00	NULL
5	C0006	Arjun	Main Road	NULL	Nagpur	Maharashtra	900.00	NULL

Define in 1 or 2 lines and give one example also

1. Recursive Relationship.

Relationship defined when a table is connected to itself, where one column acts as both the primary key and a foreign key in the same table.

2. Composite key.

Composite key is a primary key that uses two or more columns together to identify a row uniquely.

For example, in the employee(id, dept_id, name, salary) table, (id, dept_id) can form a composite key.

3. The 'like' operator with pattern matching.

The LIKE operator is used for pattern matching in SQL to find values that match a given pattern. It uses wildcard characters such as % for multiple characters and _ for a single character.

4. Drop Table command.

The DROP TABLE command is used to permanently delete a table from the database along with all its data and structure.

5. Full Outer Join.

Full Outer Join displays all records from both tables, matching rows when possible and filling NULL where no match exists.

Write queries for following descriptions: (Joins)

1. Find out the products, which have been sold to 'Ivan Bayross'.

select p.* from product_master p

join sales_order_details sod on sod.productno=p.productno

join sales_order so on so.orderno=sod.orderno

join client_master c on c.clientno=so.clientno

where c.name='Ivan Bayross';

	productno	description	profitperc	unitmeasure	qtyonhand	reorderlvl	sellprice	costprice
1	P0001	Trousers	20.00	Nos	50	10	2500.00	2000.00
2	P0002	Pull Overs	15.00	Nos	5	5	1500.00	1200.00
3	P0002	Pull Overs	15.00	Nos	5	5	1500.00	1200.00
4	P0001	Trousers	20.00	Nos	50	10	2500.00	2000.00
5	P0003	Shirts	10.00	Nos	100	20	1800.00	1400.00

2. Finding out the products and their quantities that will have to be delivered in the current month.

select p.productno,p.description,sum(qtyordered) as quant_tobe_delivered_in_this_month

from product_master p

join sales_order_details sod on sod.productno=p.productno

join sales_order so on so.orderno=sod.orderno

where month(delydate)=month(getdate()) and year(delydate)=year(getdate()) and so.orderstatus <> 'Fulfilled'

group by p.productno,p.description;

	productno	description	quant_tobe_delivered_in_this_month
1	P0001	Trousers	7
2	P0002	Pull Overs	4
3	P0003	Shirts	3
4	P0005	Socks	50

3. Listing the ProductNo and description of constantly sold (i.e. rapidly moving) products.

select p.productno,p.description from product_master p

join sales_order_details sod on sod.productno=p.productno

group by p.productno,p.description

having count(*)>2;

	productno	description
1	P0001	Trousers
2	P0002	Pull Overs

4. Finding the names of clients who have purchased 'Trousers'.

select distinct c.name from product_master p

join sales_order_details sod on sod.productno=p.productno

join sales_order so on so.orderno=sod.orderno

join client_master c on c.clientno=so.clientno

where p.description='Trousers';

	name
1	Anita
2	Ivan Bayross

5. Listing the products and orders from customers who have ordered less than 5 units of 'Pull Overs'.

select p.*, sop.* from product_master p

join sales_order_details sop on sop.productno = p.productno

where p.description = 'Pull Overs' and sop.qtyordered < 5;

	productno	description	profitperc	unitmeasure	qtyonhand	reorderlv	sellprice	costprice	orderno	productno	qtyordered	qtydisp	productrate
1	P0002	Pull Overs	15.00	Nos	5	5	1500.00	1200.00	O19001	P0002	2	2	1500.00
2	P0002	Pull Overs	15.00	Nos	5	5	1500.00	1200.00	O19004	P0002	1	1	1500.00
3	P0002	Pull Overs	15.00	Nos	5	5	1500.00	1200.00	O26002	P0002	4	0	1500.00

Write queries for following descriptions: (Subqueries)

1. Finding the non-moving products i.e. products not being sold.

select productno, description

from product_master

where productno not in (select productno from sales_order_details);

	productno	description
1	P0004	Jackets

2. Finding the name and complete address for the customer who has placed Order number 'O19001'.

```
select c.name, c.address1, c.city, c.state
from client_master c
where clientno in (
   select clientno from sales_order
   where orderno='O19001'
);
```

	name	address1	city	state
1	Ivan Bayross	A-12 MG Road	Mumbai	Maharashtra

3. Finding the clients who have placed orders before the month of May'02.

```
select c.name from client_master c
where c.clientno in(
select clientno from sales_order where orderdate<'2002-05-01'
);
```

	name
1	Suresh

Write Commands to do following

1. Display system date as Saturday, February 11, 2012

```
select format(cast('2012-02-11' as date), 'dddd, MMMM dd, yyyy');
```

	(No column name)
1	Saturday, February 11, 2012

2. Display Balance Due from Client master as $99,999.99

select format(baldue, '$#,##0.00') from client_master;

	dollar_format
1	$1,200.00
2	$800.00
3	$1,500.00
4	$500.00
5	$900.00

3. Display message as 'Salesman Aman sold goods of 50 while given target was 100.

select 'Salesman ' + salesmanname +' sold goods of ' + cast(ytdsales as varchar) + ' while given target was ' + cast(tgttoget as varchar) + '.' as message from salesman_master where salesmanname = 'Aman';

	message
1	Salesman Aman sold goods of 50.00 while given target was 100.00.

4. Display your Age in Years

select datediff(year, '02-09-2005', getdate()) as age;

	age
1	21